



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
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**EXPLORATORY STATISTICAL ANALYSIS OF CLINICAL AND
LIFESTYLE
INDICATORS FOR EARLY HEALTH RISK**

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements

for the Course of

DSA0405 – FUNDAMENTALS OF DATA SCIENCES

to the award of the degree of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**Exploratory Statistical Analysis of Clinical and Lifestyle Indicators for Early Health Risk**” has been carried out by Visha al M S (192424420), M V Raghavan (192424373), N Harshavardhan (192324189), and A Reddy Prasad (192472045) under the supervision of Dr. Krishnaraj Mony, Associate Professor / CSE, SIMATS Engineering, and is submitted in partial fulfilment of the requirements for the Bachelor of Engineering program.

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EXTERNAL EXAMINER



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DECLARATION

We, Visha al M S (192424420), M V Raghavan (192424373), N Harshavardhan (192324189), and A Reddy Prasad (192472045), of the Department of Computer Science and Engineering, SIMATS Engineering, hereby declare that the Capstone Project Work entitled “**Exploratory Statistical Analysis of Clinical and Lifestyle Indicators for Early Health Risk**” is the result of our own bonafide efforts. The project is based on the Framingham Heart Study dataset and uses descriptive statistics, Pearson correlation, significance testing and a transparent percentile-based statistical risk score without training any machine-learning or deep-learning model. We confirm that the sources, datasets and analytical procedures used in the project have been appropriately acknowledged.

Place: _____

Date: _____

Signature of the Students with Names

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INDIVIDUAL CONTRIBUTION STATEMENT

(Mandatory for all team projects)

Student / Register No.	Specific Responsibilities	Design & Analysis	Testing / Evaluation	Report Contribution	%
Vishaal M S (192424420)	Project lead; system architecture and statistical-engine design	Backend development (data ingestion, cleaning, correlation engine)	Unit testing of statistical functions; cross-schema validation	Chapters 1, 3, 4	25%
Minnal Venkatesan Raghavan (192424373)	Data collection and preprocessing design; dataset research	Data cleaning module and schema-detection logic	Dataset validation testing across multiple health datasets	Chapters 2, 3	25%
Akula Reddy Prasad (192472045)	Frontend dashboard development and visualization design	Chart.js dashboard implementation and UI development	Frontend and UI testing	Chapters 3, 4	25%
Nadavadi Harshavardhan (192324189)	Documentation, literature review, and report compilation	Literature review and reference compilation	Documentation review and consistency checking	Chapters 2, 5, References	25%
Total					100

ABSTRACT

This project focuses on exploratory statistical analysis of clinical and lifestyle indicators to assess early health risk. It uses the Framingham Heart Study dataset to compute descriptive statistics, Pearson correlation, significance testing and a transparent percentile-based statistical risk score. The project addresses a gap in traditional health screening, where single-indicator thresholds or manual visual inspection may not adequately examine relationships among multiple clinical and lifestyle indicators.

The proposed system is a statistically grounded, dashboard-based analytical platform. It automatically ingests the health dataset, detects the schema and binary outcome column, encodes categorical fields, filters identifier columns, performs missing-value imputation, classifies feature types, validates the dataset, calculates correlation coefficients and p-values, computes a transparent risk score, and presents statistical results through an interactive dashboard.

The methodology uses Python, Pandas, NumPy, SciPy, FastAPI and Chart.js. The principal techniques are descriptive statistics, Pearson correlation analysis, significance testing using p-values, percentile-based risk scoring, median/mode imputation, categorical encoding and outcome-group comparison using a t-test. The risk-scoring component uses correlation-weighted Z-scores followed by percentile ranking against the study population. No machine-learning or deep-learning model is trained.

The system is organized into four modules: Data Ingestion & Schema Detection; Data Cleaning & Preprocessing; Statistical Correlation & Risk Scoring; and Visualization & Reporting. The intended result is a transparent analytical view of which indicators are most strongly associated with the selected health-risk outcome and an interpretable relative risk position for an individual compared with the study population.

The PPT identifies future enhancements including live clinical data feeds, additional statistical inference techniques, improved outcome-detection heuristics, expanded dashboard comparison views, large-scale dataset optimization and deployment in preventive health-screening programs.

Keywords: Exploratory Statistical Analysis; Clinical Indicators; Lifestyle Indicators; Early Health Risk; Framingham Heart Study; Pearson Correlation; Descriptive Statistics; Statistical Significance; Percentile-Based Risk Score

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The PPT contains the architecture and module descriptions but not final editable Word figures. Insert final implementation diagrams/screenshots in these locations.

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Meaning / Unit
CSE	Computer Science and Engineering	Department
DSA	Data Sciences	Course context
SciPy	Scientific Python	Statistical library
API	Application Programming Interface	Software interface
Z-score	Standardized score	Standard deviations
p-value	Statistical significance value	Probability value
ML	Machine Learning	Not trained in this project
DL	Deep Learning	Not trained in this project
BMI	Body Mass Index	Health indicator if present
Kaggle	Data platform	Dataset source identified in PPT

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

The PPT states that this project focuses on exploratory statistical analysis of clinical and lifestyle indicators to assess early health risk. It uses the Framingham Heart Study dataset to compute descriptive statistics, Pearson correlation and a transparent, percentile-based statistical risk score.

Traditional screening may rely on single-indicator thresholds or manual visual inspection. The project instead examines relationships among multiple risk indicators and provides statistically grounded, explainable outputs without training an ML or DL model.

1.2 Need for the Project

The identified gap is between routine health-data collection and rigorous statistical verification. A dashboard-based system can automatically summarize indicators, identify associations, test statistical significance and report an individual's relative position against the study population.

1.3 Problem Statement

Traditional health-screening methods rely on single-indicator thresholds or manual visual inspection and do not proactively evaluate correlations between multiple risk indicators. This creates a critical gap between routine data collection and rigorous statistical verification.

1.4 Project Aim

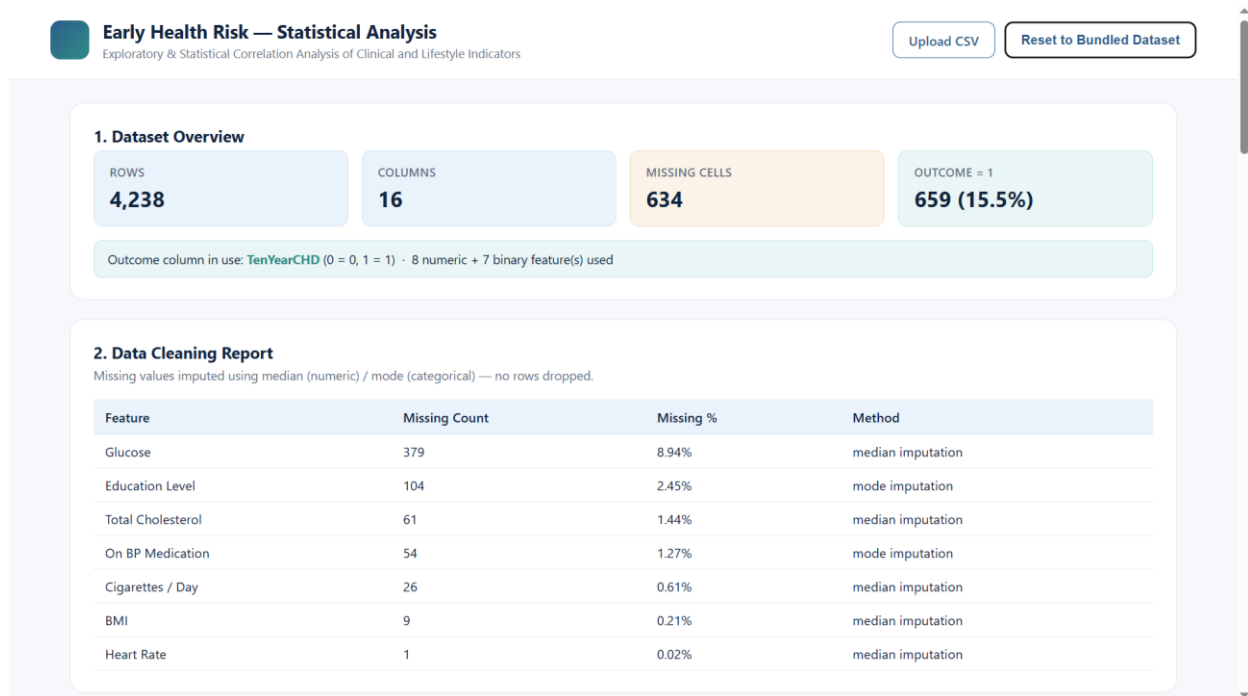
To design a statistically grounded, dashboard-based system that identifies clinical and demographic factors associated with early health risk and reports a transparent, explainable risk score.

1.5 Project Objectives

1. Identify clinical and demographic factors most strongly associated with early health risk using Pearson correlation.
2. Automatically compute mean, median, standard deviation and range for clinical and lifestyle indicators.
3. Generate histograms, correlation charts and other analytical visualizations.
4. Provide a transparent statistical risk score describing an individual's relative risk position against the study population.

1.6 Scope

The scope covers the Framingham Heart Study dataset, clinical and lifestyle indicators, descriptive statistics, Pearson correlation, p-value significance testing, median/mode imputation, categorical encoding, t-test outcome-group comparison, correlation-weighted Z-score scoring, percentile ranking and dashboard reporting.



The project explicitly does not train a machine-learning or deep-learning model. The statistical score is a relative analytical measure and should not be treated as a clinical diagnosis.

1.7 Expected Engineering Outcome

The expected outcome is an automated and transparent statistical-analysis platform that converts a health dataset into descriptive summaries, statistically tested correlations, outcome-group comparisons, visualizations and a percentile-based relative risk score.

1.1 Proposed System Architecture

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The literature review follows the five research-paper entries provided in the PPT and considers statistical foundations, predictive analytics, data mining, lifestyle-based scoring and statistical modelling for health-risk assessment.

2.2 Literature Review

S.No.	Research Paper	Author & Year	Advantages	Disadvantages	Technologies
1	Framingham Study Insights: Statistical Foundations of Cardiovascular Risk Assessment	Kannel & McGee (2023)	Established statistical basis; large longitudinal dataset	Older population sample; limited risk factors	Statistical Analysis, Python, Epidemiology
2	Predictive Analytics for Chronic Disease Risk Assessment Using Clinical Data	Sharma & Iyer (2024)	Identifies key risk indicators; interpretable	Requires clean, complete clinical data	Statistical Regression, Python, Pandas
3	Data Mining Approaches for Early Detection of Cardiovascular Risk Factors	Menon et al. (2024)	Surfaces non-obvious patterns	Computationally intensive at large scale	Data Mining, Decision Trees, Python
4	A Data-Driven Framework for	Nair & Das (2025)	Combines clinical and	Sensitive to self-reported	Data Analytics,

	Lifestyle-Based Health Risk Scoring		lifestyle factors; transparent	lifestyle accuracy	Correlation Analysis
5	Statistical Modeling for Early Diabetes Risk Prediction Using Health Records	Krishnan et al. (2025)	Strong statistical grounding; explainable factors	Domain-specific; needs adaptation	Logistic Regression, Statistical Modeling

2.3 Comparative Analysis

The reviewed approaches range from classical statistical analysis to regression and data mining. The proposed project is intentionally positioned toward transparent exploratory statistics and does not train an ML/DL prediction model.

2.4 Research / Engineering Gap

The project addresses the need for a simple, explainable and dashboard-oriented workflow combining descriptive statistics, correlation analysis, significance testing and transparent risk scoring.

2.5 Proposed Contribution

The contribution is an integrated workflow that automatically detects the outcome field, cleans the dataset, calculates descriptive and inferential statistics, derives a correlation-weighted score, ranks the score against the study population and reports results visually.

1.2 Statistical Risk Scoring Workflow



CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder	Requirement
-------------	-------------

Student / Analyst	Automated cleaning, descriptive and correlation analysis.
Screening User	Understand relative statistical risk position.
Supervisor / Evaluator	Verify transparent and reproducible statistical methods.
Data Administrator	Protect uploaded health datasets.
Future Health Program	Use aggregated insights responsibly.

Type	Requirement
Functional	Detect dataset schema and binary outcome column.
Functional	Encode categorical fields and filter identifiers.
Functional	Impute missing values using median/mode.
Functional	Compute descriptive statistics.
Functional	Compute Pearson correlation and p-values.
Functional	Compute percentile-based statistical risk score.
Functional	Display histograms, correlation and comparison plots.
Non-Functional	Transparent and traceable calculations.
Non-Functional	Controlled access to sensitive data.
Non-Functional	Usable dashboard.

3.2 Constraints & Applicable Standards

Constraint / Practice	Application
Statistical validity	Use descriptive statistics, Pearson correlation, p-values and t-test as specified.
Data quality	Median/mode imputation and consistency validation.
Privacy	Minimize exposure of personal health information.

Explainability	Correlation-weighted Z-score and percentile ranking.
Software engineering	Four modular components with separated responsibilities.

3.3 Design Specifications

Parameter	Requirement	Output	Priority
Dataset ingestion	Detect schema and outcome	Validated dataset	High
Preprocessing	Handle missing values and types	Cleaned dataset	High
Descriptive statistics	Mean, median, SD, range	Statistical summary	High
Correlation	Pearson coefficient	r values	High
Significance	p-value testing	p values	High
Risk score	Correlation-weighted Z-score + percentile	Risk percentile	High
Visualization	Histograms and correlation charts	Dashboard	Medium
Backend	FastAPI interface	API	Medium

3.4 Alternative Solutions & Evaluation

Alternative	Advantages	Limitations	Decision
Manual spreadsheet	Simple for small data	Slow and less reproducible	Not primary
Statistical scripting	Transparent and reproducible	Requires programming	Selected

Machine learning	Can model complex patterns	Outside stated scope; less transparent	Not selected
Deep learning	Can learn complex patterns	High complexity; outside scope	Not selected
Dashboard statistics	Automated and visual	Requires application development	Selected

3.5 Selected Design & Detailed Design

The selected design follows the four modules explicitly presented in the PPT.

DATA INGESTION & SCHEMA DETECTION → DATA CLEANING & PREPROCESSING → STATISTICAL CORRELATION & RISK SCORING → VISUALIZATION & REPORTING

Module 1 – Data Ingestion & Schema Detection: ingests the uploaded health dataset, detects the binary outcome column, encodes categorical fields and filters identifier columns. Techniques include name-based and structural target detection, label encoding/factorization and uniqueness-based identifier filtering.

Module 2 – Data Cleaning & Preprocessing: handles missing values, classifies feature types and validates the dataset. Numeric fields use median imputation and categorical fields use mode imputation.

Module 3 – Statistical Correlation & Risk Scoring: calculates Pearson correlation coefficients, p-values, correlation-weighted Z-scores and percentile rankings against the study population.

Module 4 – Visualization & Reporting: presents statistical data, outcome-group comparisons, correlation charts and a transparent risk-score breakdown using Chart.js.

3.6 Trade-off Analysis

Decision	Alternative	Benefit	Trade-off	Final Decision
Statistical score	Opaque predictive model	Explainable	May miss complex nonlinear relationships	Selected
Median/mode imputation	Delete incomplete rows	Retains data	Adds assumptions	Selected

Correlation analysis	Complex feature learning	Interpretable	Association is not causation	Selected
Percentile score	Absolute threshold	Relative position	Not a clinical threshold	Selected
Dashboard	Static tables	Interactive communication	UI effort	Selected

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Action
Sustainability	Avoid unnecessary computational complexity	Use lightweight statistical methods appropriate to exploratory analysis.
Ethics	Health data is sensitive	Minimize exposure and control access.
Safety	Score could be misunderstood	Clearly label it as relative statistical screening, not diagnosis.
Fairness	Associations may vary by population	Document limitations and avoid causal claims.
Security	Uploaded datasets require protection	Apply authentication/access control and secure storage in deployment.
Reproducibility	Outputs must be traceable	Document preprocessing and scoring steps.

Risk	Likelihood	Severity	Level	Mitigation	Residual
Missing/inconsistent data	High	Medium	High	Imputation and validation	Medium
Incorrect outcome detection	Medium	High	High	Structural/name checks and verification	Medium

Misinterpretation as diagnosis	Medium	High	High	Clear labels and human oversight	Low-Medium
Privacy exposure	Low-Medium	High	High	Access control and anonymization	Low-Medium
Correlation treated as causation	Medium	Medium	Medium	Explicit interpretation limits	Low
Dashboard calculation error	Low-Medium	Medium	Medium	Independent test calculations	Low

1.3 Implementation Workflow

t = -16.436, p = 0 — statistically significant difference ($\alpha = 0.05$)

Descriptive Statistics

Feature	Mean	Median	Std Dev	Min	Max
Age	49.66	49	8.38	32	70
Cigarettes / Day	9.42	0	12.93	0	70
Total Cholesterol	199.09	199.6	34.79	107	366.1
Systolic BP	123.09	123	18.53	83	189
Diastolic BP	76.35	76.2	13.2	48	123.9
BMI	25.52	25.49	4.02	15	42.48
Heart Rate	76.92	77	12.17	44	120
Glucose	79.86	80	15.69	40	145
Male	0.42	0	0.49	0	1
Education Level	1.96	2	1.02	1	4
Current Smoker	0.5	0	0.5	0	1
On BP Medication	0.02	0	0.14	0	1
History of Stroke	0	0	0.07	0	1
Hypertensive	0.16	0	0.37	0	1
Diabetes	0.02	0	0.16	0	1

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The PPT identifies Python, Pandas, NumPy, SciPy, FastAPI and Chart.js as the principal technologies. The Framingham Heart Study dataset is identified as the main dataset, with health data sources including a Kaggle health dataset, clinical records and lifestyle surveys.

Tool / Resource	Purpose
Python	Programming and analysis
Pandas	Data manipulation
NumPy	Numerical computation
SciPy statistics	Correlation and significance testing
FastAPI	Backend/API layer
Chart.js	Interactive visualization
Framingham Heart Study dataset	Primary dataset
Kaggle health dataset	Health-data source identified in PPT
Clinical records	Potential data source
Lifestyle surveys	Potential data source

4.2 Implementation Workflow

DATA INGESTION → SCHEMA DETECTION → ENCODING / ID FILTERING →
IMPUTATION → FEATURE CLASSIFICATION → DESCRIPTIVE STATISTICS → PEARSON
CORRELATION + p-VALUE → CORRELATION-WEIGHTED Z-SCORE → PERCENTILE
RANKING → DASHBOARD

The implementation produces an analysis-ready dataset, encoded feature set, descriptive statistics, correlation coefficients, p-values, statistical risk percentile, graphs and a transparent score breakdown.

4.3 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion	Status
Outcome detection	Run representative datasets	Correct binary outcome detected	To verify

Encoding	Inspect categorical fields	Consistent encoding	To verify
Missing values	Check/inject missing values	Handled correctly	To verify
Descriptive statistics	Independent calculation	Values agree	To verify
Pearson correlation	Independent calculation	Coefficient agrees	To verify
p-value	Independent statistical test	Value agrees	To verify
Risk score	Manual sample calculation	Score/percentile agrees	To verify
Dashboard	Functional test	Charts and score display correctly	To verify

4.4 Results

The supplied PPT describes the analysis workflow, modules and intended outputs but does not provide final numerical experimental results in the extracted slide content. Therefore, no unsupported correlation values, p-values, risk scores or accuracy values are invented in this report.

Result Area	Expected Output
Data processing	Validated and cleaned health dataset
Descriptive analysis	Mean, median, standard deviation and range
Correlation	Pearson coefficients and p-values
Outcome comparison	t-test comparison plots/results
Risk scoring	Correlation-weighted Z-score and percentile
Visualization	Histograms, correlation charts and dashboard

4.5 Data Analysis & Engineering Judgment

Pearson correlation provides an interpretable measure of linear association, while p-values support significance testing. The risk score standardizes indicator values using Z-scores, incorporates correlation information and ranks the resulting score against the study population. Statistical

association should not be interpreted as causation, and the percentile score should not be treated as a medical diagnosis.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence in PPT	Achievement
Identify associated factors	Pearson correlation + p-value	Specified / implemented
Compute descriptive statistics	Mean, median, SD, range	Specified / implemented
Generate visualizations	Histograms and correlation charts	Specified / implemented
Provide transparent score	Correlation-weighted Z-score + percentile	Specified / implemented
Avoid ML/DL training	Explicitly stated in PPT	Achieved by design

4.7 Strengths & Limitations

Strengths

- Transparent statistical methodology.
- No trained ML/DL model required.
- Automatic schema detection and preprocessing.
- Correlation and p-values are interpretable.
- Percentile score shows relative population position.
- Dashboard supports visual interpretation.

Limitations

- Pearson correlation captures linear association only.
- Correlation does not establish causation.
- Percentile score is not a clinical threshold.
- Self-reported lifestyle data may be inaccurate.
- Outcome-detection heuristics require validation.
- The PPT does not report final numerical experimental results.

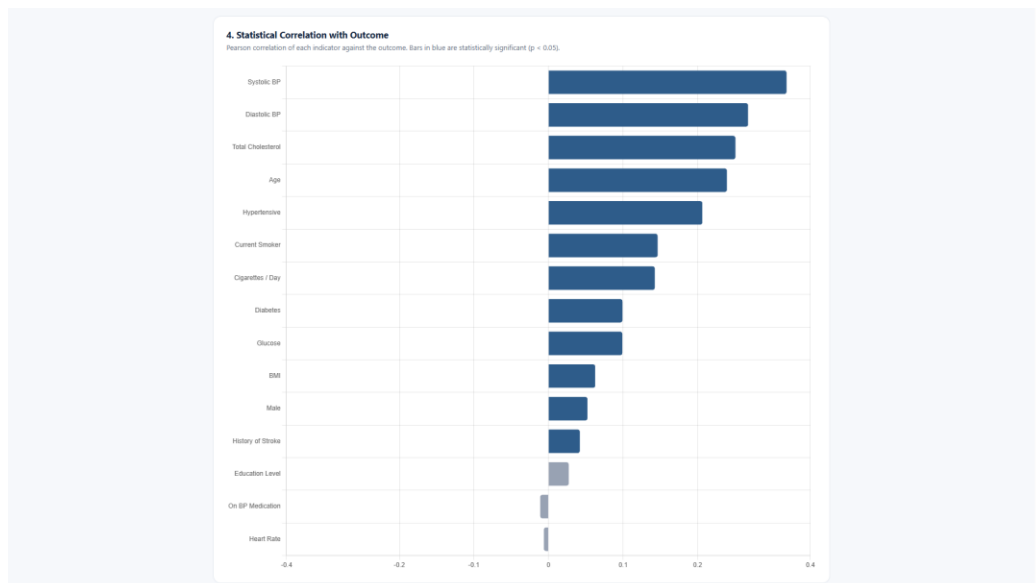
4.8 Project Planning, Roles & Budget

Milestone	Activity	Status
-----------	----------	--------

1	Problem definition and objectives	Completed in PPT
2	Literature review	Completed in PPT
3	Data ingestion and preprocessing	Defined
4	Statistical analysis and risk scoring	Defined
5	Dashboard/reporting	Defined
6	Testing and validation	To verify
7	Final documentation	To complete

Budget Item	Estimated ₹	Actual ₹
Python / Pandas / NumPy / SciPy	0	0
FastAPI / Chart.js	0	0
Dataset	To verify	To verify
Cloud / deployment	To determine	To determine
Printing / documentation	To determine	To determine
Total	To determine	To determine

1.4 Visualization / Dashboard Layout



CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The project demonstrates how exploratory statistical analysis and correlation-based methods can improve early health-risk assessment. It enables transparent, evidence-based risk screening without relying on any trained machine-learning or deep-learning model. The four-module architecture creates a structured path from raw health data to descriptive statistics, correlations, significance testing, relative risk scoring and dashboard reporting.

5.2 Future Work

5. Integration with live clinical data feeds.
6. Additional statistical inference techniques.
7. Improved outcome-detection heuristics.
8. Expanded dashboard comparison views.
9. Large-scale dataset optimization.
10. Deployment in preventive health-screening programs.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- Descriptive statistical analysis.
- Pearson correlation and p-value testing.
- Median/mode imputation and feature classification.
- Correlation-weighted Z-score scoring.
- Percentile ranking.
- Dashboard-oriented statistical reporting using FastAPI and Chart.js.

Engineering and Professional Skills

- Problem decomposition and modular design.
- Data cleaning and validation.
- Transparent statistical reasoning.
- Interpretation of statistical associations.
- Technical documentation and visualization.

- Responsible handling of sensitive health information.

Individual Reflection – A Reddy Prasad (192472045)

The preprocessing workflow highlighted the importance of missing-value handling, feature-type classification and identifier filtering. Modular preprocessing improves reproducibility and reduces invalid statistical operations.

CHAPTER-6

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CHAPTER-7

APPENDICES

Appendix A – Detailed Statistical Workflow

Dataset → Outcome Detection → Categorical Encoding → Identifier Filtering → Missing-Value Imputation → Feature-Type Classification → Descriptive Statistics → Pearson Correlation → p-value Testing → Correlation-Weighted Z-score → Percentile Ranking → Visualization and Reporting.

Appendix B – Module Summary

Module	Purpose	Core Functions	Output
1. Data Ingestion & Schema Detection	Prepare uploaded dataset	Outcome detection; encoding; ID filtering	Validated dataset; encoded features
2. Data Cleaning & Preprocessing	Prepare data for analysis	Imputation; feature classification; validation	Cleaned dataset
3. Statistical Correlation & Risk Scoring	Find associations and score	Pearson r; p-values; Z-score; percentile	Correlation, significance and risk percentile
4. Visualization & Reporting	Communicate results	Charts; group comparison; score reporting	Dashboard and graphs

Appendix C – Statistical Formulae

Z-score: $Z = (X - \mu) / \sigma$.

Pearson correlation: $r = \text{Cov}(X,Y) / (\sigma_X \sigma_Y)$.

The project uses correlation-weighted Z-scores followed by percentile ranking. The exact production formula should be reproduced from the final source code before submission.

Appendix D – Data Sources

Source	Use
Framingham Heart Study dataset	Primary dataset identified in PPT
Kaggle health dataset	Health-data source identified in PPT

Clinical records	Potential source identified in PPT
Lifestyle surveys	Potential source identified in PPT

Appendix E – Dashboard Evidence

Insert final screenshots of histograms, correlation charts, outcome-group comparison plots and the transparent risk-score breakdown here. The PPT describes these outputs but does not contain final editable dashboard screenshots.

Appendix F – Ethics / Privacy Checklist

- Use health data only for the stated analytical purpose.
- Avoid unnecessary personally identifiable information.
- Apply access control to uploaded datasets.
- Anonymize/de-identify records where appropriate.
- Clearly distinguish statistical screening from medical diagnosis.
- Document population and methodological limitations.

Appendix G – Project Evidence

Attach implementation screenshots, test records, statistical-output verification, version-control evidence and supervisor review records where available.

Appendix H – Outcome Mapping

Outcome Area	Evidence
Problem formulation	Chapter 1
Engineering design	Chapter 3 requirements, architecture and trade-offs
Statistical analysis	Chapter 4 methods and evaluation
Ethics / responsibility	Chapter 3 safety, privacy, risk and security
Teamwork	Contribution statement and planning
Communication	Dashboard and technical report
Independent learning	Chapter 5 reflection

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8) + Appendix J